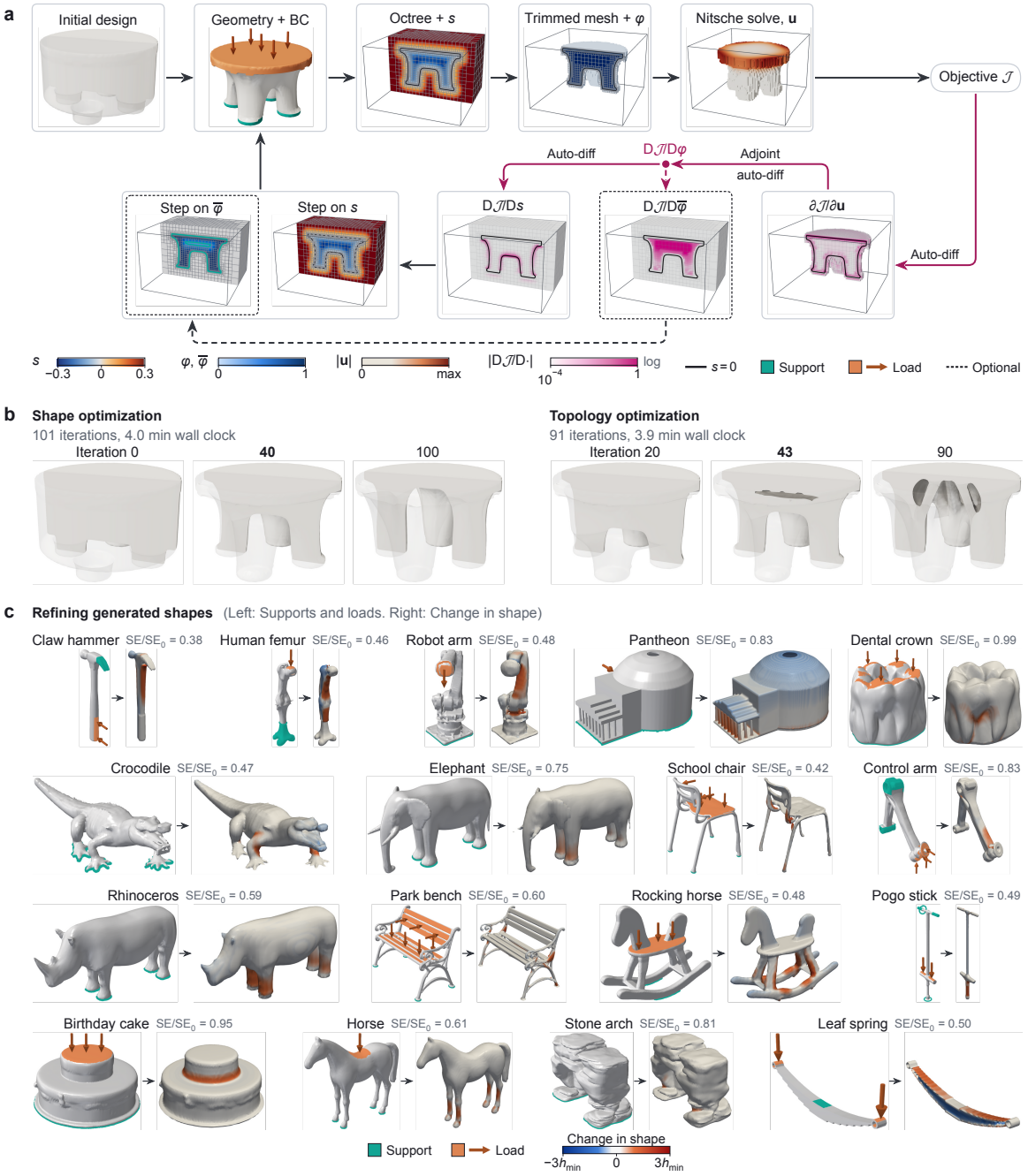

a
Initial design
Geometry + BC
Octree + s
Trimmed mesh + φ
Nitsche solve, u
Objective 𝒥
Auto-diff
D𝒥/Dφ
Adjoint
auto-diff
Step on φ̄
Step on s
D𝒥/Ds
D𝒥/Dφ̄
∂𝒥/∂u
Auto-diff
s
−0.3
0
0.3
φ, φ̄
0
1
|u|
0
max
|D𝒥/D·|
log
10^−4
1
s = 0
Support
Load
Optional
b
Shape optimization
101 iterations, 4.0 min wall clock
Iteration 0
40
100
Topology optimization
91 iterations, 3.9 min wall clock
Iteration 20
43
90
c
Refining generated shapes
(Left: Supports and loads. Right: Change in shape)
Claw hammer SE/SE0 = 0.38
Human femur SE/SE0 = 0.46
Robot arm SE/SE0 = 0.48
Pantheon SE/SE0 = 0.83
Dental crown SE/SE0 = 0.99
Crocodile SE/SE0 = 0.47
Elephant SE/SE0 = 0.75
School chair SE/SE0 = 0.42
Control arm SE/SE0 = 0.83
Rhinoceros SE/SE0 = 0.59
Park bench SE/SE0 = 0.60
Rocking horse SE/SE0 = 0.48
Pogo stick SE/SE0 = 0.49
Birthday cake SE/SE0 = 0.95
Horse SE/SE0 = 0.61
Stone arch SE/SE0 = 0.81
Leaf spring SE/SE0 = 0.50
Support
Load
Change in shape
−3h_min
0
3h_min